SE | SAE | Operator | QKD / KME | Peer / origin SAE (RMI) | End without keys

Pair locks
Every operation on a client pair (new share, stored share, operator delete) locks the pair on both SAEs: the lower SAE ID first, then the higher. A refusal releases the first lock; nobody waits.
Lease: in memory, 5 min, random token and issuing SAE; renewed every 60 s and before and after each commit step.
Fencing: every write and delete checks in one transaction that the lease is still held. A lost lock aborts the operation and closes the device session.
Release: remote first.
Stored share: one state, keys available, with the lock token, the issuer and the counterpart SAE.
RMI: mutual TLS with pinned peers; the caller is identified by its certificate.

PROVISION with current time
PIN stays off USB

Device ready to provision?
owner, device certificate + key,
SAE CA, ML-KEM key;
time synced

No → Refused: TLS is not armed

Yes

Post-quantum mutual TLS 1.3
Device checks the SAE certificate
SAE checks the device certificate

Device sends: session signature, its identity,
slot size and pad count, ML-KEM public key,
its peers (identity, name, pad size)

in parallel → Device erases its QKD key slots,
then reports wipe finished

Uplink from this device
and this TLS session?
(identity hash and signature)

No → Abort (bad uplink)

Yes

Operator: choose peer SAE and peer device;
pad size = smaller of the two devices

No: Reselect

Stored share for this pair?

No

Yes, other SAE

Yes, same target

No shared keys were found for this pair.
Generate new keys now?
(skipped after a delete)

Share with the selected SAE:
created by this SAE?

Shared keys already exist
with SAE … (created …).
Did you mean to use
those keys?

No → This will permanently delete
those keys. Do you really
want to delete them and
generate new ones?

No → Existing keys were kept.
No keys were sent to the device.

Yes → Keys created by the other SAE
and the device has room?

No → Keys generated on this SAE, or the
chip is smaller. No keys sent.

Yes → Lock both SAEs (lower ID first);
local share is the one shown;
read the share on the origin

Lock refused, origin unreachable or
share changed. Keys were not sent.

Use-time check: both SAEs hold
the same share, naming each other?

Yes → Read the keys; delete the share
on the origin, then locally

No → Delete the orphan, remote first.
The stored keys were deleted and not
sent. Provision both devices again.

Yes → ask again

Yes

Lock the pair: lower SAE ID first,
then higher

refused → Another operation for this device pair
is in progress on an SAE. Retry later.

both held

Still no local share.
Does the target SAE hold
a share with another SAE?

Yes → Release the locks.
This will permanently delete
those keys. Delete them and
generate new ones?

No → Existing keys were kept.
No keys were sent to the device.

Yes (on the target)

No, or names this SAE

Generating new keys.
Get keys from the KME,
then renew both locks

deleted: restart,
prompt skipped

Shared keys already exist with
SAE … (created …).
Delete them and generate
new keys?

No → Existing keys were kept.
No keys were sent to the device.

Yes → Operator delete: lock both SAEs
(lower ID first), re-read the shown share;
delete only if unchanged:
counterpart first, then the shown share;
release remote first

share changed → The shared keys changed while you were
deciding. Nothing was deleted; check them again.

Every grey end closes TLS without a downlink. The device has already erased its QKD key slots after the uplink. Errors (QKD, RMI) end the session with an operator error message. Every end releases the pair locks, remote first.

Build the downlink: ML-KEM encapsulation,
sealed pads for the origin device

Build the downlink: ML-KEM encapsulation,
sealed pads for the peer device

Wait for wipe finished, then
stream the pads to the device
(progress every 10 KiB);
renew both locks every 60 s

lock lost or timeout → Abort / close TLS

Device stores all pads?

No → Abort / close TLS

Yes

New share generated?

Yes → Renew both locks; store the share
on the peer, then on the origin;
renew both locks

No (stored keys were used)

Both rows written,
locks still held?

No → Peer share deleted. Device stored
the keys, but the share could not be
recorded on both SAEs. Provision both
devices again.

Yes → Loading to device: 100 %
Share successful.

Release both locks, remote first

wipe finished